

Australian stick insects turn out to be five species, not two

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How can a large, often photographed insect still surprise scientists? In Australia, some stick insects looked familiar, but their identities needed another careful check.

A University of Sydney team studied Australian insects in the group called Anchiale. Insects previously classified as two **species** turned out to represent five separate **species**. They live from northern New South Wales along coastal Queensland and into tropical northern Australia.

The team identified two previously unrecognized **species** and restored recognition of a third. One large insect had been incorrectly identified for decades. It is now recognized as its own **species**, Anchiale robusta.

The researchers gathered several kinds of clues. They studied **DNA**, the material that carries inherited information, alongside museum **specimens**, field surveys and insects raised in **captivity**. They also used observations shared by members of the public.

One newly described **species**, Anchiale mabiensis, lives in Queensland's critically endangered Mabi rainforest. Its eggs have an unusual lid structure that had not previously been recorded in stick insects or their closest relatives.

Knowing which **species** is which can matter outside a museum. Some stick insects sometimes become so numerous that they strip leaves from trees. One insect associated with these **outbreaks** turned out to include more than one **species**. Telling them apart could help people understand effects on forests and farming.

The team also suspects another new **species** lives in Cape York. More **specimens** are needed before that possibility can be **confirmed**.

Word Treasure

species -- A distinct kind of living thing.

DNA -- Material in living things that carries inherited information.

specimens -- Examples of animals or other living things collected for study.

captivity -- Being kept and cared for by people instead of living in the wild.

outbreaks -- Periods when the number of these insects becomes unusually large.

confirmed -- Checked and shown to be correct.

Background you need

DNA carries inherited information. In this study, researchers examined it alongside specimens and observations.

Scientists can discover that a familiar animal belongs to a species that was not previously recognized.

Break it down

Opens with a question about how familiar insects can surprise scientists.

Explains the change from two recognized species to five.

Describes the evidence and two named species.

Explains possible benefits for forests and farming.

Ends with a possible species that needs further study.

Why it matters

Identifying species more accurately can reveal overlooked insect diversity and could help people understand effects on forests and farming.

Test what you remember

1. How many separate species did the team recognize among insects previously classified as two species?
 - (A) Four species
 - (B) Six species
 - (C) Five species
 - (D) Three species
2. Which university's team studied the stick insects?
 - (A) The University of Melbourne
 - (B) The University of Queensland
 - (C) The University of Adelaide
 - (D) The University of Sydney
3. What happened to the insect called *Anchiale robusta*?
 - (A) It was moved to another rainforest.
 - (B) It was recognized as a separate species.
 - (C) It was confirmed to live in Cape York.
 - (D) It was found only in museum collections.
4. What is unusual about the eggs of *Anchiale mabiensis*?
 - (A) They have a distinctive lid structure.
 - (B) They have a bright striped pattern.
 - (C) They have a soft feathery covering.
 - (D) They have a smooth square shape.
5. Why could telling the outbreak-related species apart be useful?
 - (A) It could explain changes in ocean water.
 - (B) It could help explain effects on forests.
 - (C) It could reveal how birds choose nests.
 - (D) It could help plan new museum buildings.
6. What does the team need to confirm the possible new species in Cape York?
 - (A) More photographs of nearby trees
 - (B) More records of farming methods
 - (C) More specimens of the insects
 - (D) More maps of rainforest paths

Answer key (try the quiz first!): 1: C 2: D 3: B 4: A 5: B 6: C

MY THOUGHTS (at least 20 words):